

LUCY Boundary Threshold Framework

Professional reading edition of the evolving public MKUFT canon

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Abstract

LUCY — **Local Unified Coherence Yield** — is MKUFT's threshold architecture for the point at which an addressed relational or informational configuration becomes sufficiently coherent and closed to produce a consequential change in the next addressed layer. In the physics-facing case, the central object is the crossing at which an I-layer relation becomes physically consequential at P by changing a declared physical transition, admissibility, weighting, boundary condition, or effective state.

LUCY is therefore **not intrinsically a bubble, circle, membrane, pocket, or spatial shape**. Those can be downstream physical morphologies when a particular implementation produces them. The primary invariant is the crossing condition itself.

Where a stable interface and finite region do emerge, the useful downstream sequence remains **LUCY crossing threshold → LUCY-1 membrane → LUCY-2 sustained pocket**. LUCY-1 and LUCY-2 are conditional physical regimes, not compulsory geometry in every LUCY event.

The framework distinguishes **coherence** from the stronger condition of **relational closure**. Coherence means that active components or constraints are mutually consistent enough to reinforce a candidate state. Relational closure means that the relevant constraint loop is sufficiently complete for the state to persist, resist perturbation, or lawfully alter downstream routing.

LUCY does not claim that gravitation, inertia, light propagation, nuclear stability, cognition, or information routing are identical. It is a typed threshold architecture that may be addressed separately at informational, physical, and observer layers. Evidence cannot be inherited between those addresses without a defined carrier.

For physical applications, LUCY does not propose that a gravitational wave, particle, or force is amplified. Its stronger open hypothesis is that a relational state may cross a threshold at which its structure becomes physically consequential. A stable boundary or different effective physical geometry may follow in some regimes, but neither is built into the definition of LUCY itself.

The proposal overlaps established work on phase transitions, domain walls, emergent spacetime, induced gravity, analogue gravity, scalarisation, and collective states. Those topics are not claimed as original. The proposed MKUFT contribution is the typed conjunction of layer addressing, a cross-layer crossing condition, relational closure, optional membrane and sustained-pocket regimes, discriminating physical readout, strongest-fair-null testing, ordinary-physics recovery, and explicit falsification.

1. Exact object

LUCY means **Local Unified Coherence Yield**.

The exact proposal is:

A declared relational state reaches a prespecified coherence or closure threshold at which it becomes consequential at the next addressed layer.

For the physics-facing I→P case:

an informational or relational configuration earns a physical LUCY claim only when a defined coupling changes a measurable P-layer transition, admissibility, weighting, boundary condition, or effective state beyond what the adequate physical baseline already predicts.

A stable boundary and sustained finite region are possible downstream consequences, not part of the minimum definition.

Every application requires an addressed source layer, target layer, state variables, coherence or closure measure, control parameters, threshold, typed coupling or carrier, predicted post-threshold behaviour, ordinary baseline, and falsifier. A boundary condition is additionally required when the application specifically claims a membrane or pocket. Without those items LUCY remains descriptive language.

2. Crossing condition and downstream regimes

2.1 LUCY — crossing threshold

LUCY is the critical condition at which a candidate relational state becomes sufficiently coherent, closed, or otherwise well-defined to alter a declared downstream relation.

Below threshold, local alignment or fluctuation may occur without a consequential change at the target layer. At threshold, the addressed configuration becomes capable of changing the target-layer dynamics under the declared coupling.

In a physical application this means that the LUCY claim resides in the **I→P handshake**, not in any assumed visible shape. A membrane, domain, phase boundary, altered path weighting, state transition, or other physical morphology may follow according to the specific P-layer model.

2.2 LUCY-1 — membrane, when present

A **LUCY membrane** is a stable interface separating two differently resolved local conditions. “Membrane” is functional language: the interface regulates crossing, coupling, routing, or state continuity. It does not require a biological membrane or literal sheet of matter.

LUCY-1 is a downstream morphology-specific regime. A valid LUCY crossing does not require LUCY-1 unless the active model predicts a stable interface.

A physical membrane claim should specify boundary thickness or resolution limit, lifetime, permeability or coupling rule, surface energy or maintenance cost, perturbation response, and observables locating the boundary.

2.3 LUCY-2 — sustained pocket, when present

LUCY-2 is a finite region maintained by a LUCY-1 boundary. The preferred term is **sustained pocket** or **sustained local region**. “LUCY field” is appropriate only where a field is mathematically defined in the active domain.

LUCY-2 is therefore conditional on a boundary-supporting implementation. It is not the generic geometry of LUCY.

A sustained pocket requires more than a momentary boundary. It must retain a distinguishable internal state for a declared time under declared noise, loading, and environmental conditions.

3. Coherence and relational closure

3.1 Coherence

Coherence is alignment or mutual consistency among the active constraints, phases, routes, or state variables relevant to the addressed system. Operational indicators may include reduced phase dispersion, reduced routing conflict, increased correlation, stable ordering, reduced transition noise, or persistent low-cost traversal.

3.2 Relational closure

Relational closure is achieved when the active relations form a sufficiently complete loop to maintain the state or lawfully constrain its downstream expression.

The candidate causal grammar is:

state → downstream response → correction, reinforcement, or constraint → retained or resolved state.

Where a membrane exists, that loop may include boundary maintenance. Where no membrane exists, closure may instead appear as stable routing, altered admissibility, or another declared target-layer consequence.

Closure adds persistence, reciprocal constraint, perturbation correction, and a sufficiently complete action–response loop. The stronger physical LUCY claim begins at consequential closure rather than resemblance, synchrony, or high correlation alone.

3.3 Deeper constraint-closure hypothesis

Coherence may be an observable consequence of a deeper transition in which a network of constraints becomes mutually satisfiable, reciprocally supporting, and sufficiently closed to persist under perturbation or alter downstream admissibility.

The proposed cross-domain grammar is:

constraint compatibility → reciprocal support → closure threshold → consequential layer crossing → domain-specific resolved behaviour.

For boundary-forming physical branches this can continue as:

consequential layer crossing → stable boundary → sustained local regime.

This does not assert that nuclear stability, collective physical phases, biological organisation, cognition, and effective geometry share one mechanism or equation. It proposes a common structural question: whether a previously unstable set of relations becomes self-supporting enough to change what the next addressed layer can stably do.

The hypothesis is retained only if it adds predictive discrimination beyond established domain models. It fails as a physical addition when “constraint closure” merely redescribes stability after the fact, cannot be operationalised independently, or transfers mechanism between domains without a defined carrier.

3.4 O-layer closure address

Within the recursive MKUFT architecture, *O* is the address at which a realised configuration becomes registered as a sufficiently determinate relation to complete a loop and become available for recurrence.

The source-level cycle is

$$S \rightarrow I \rightarrow P \rightarrow O \rightarrow S.$$

Applied locally to a physics-facing LUCY crossing, the proposed sequence is **constraint compatibility at I → consequential physical realisation at P → registration and closure at O → updated admissibility for later states**.

Where the physical realisation produces a stable boundary, the sequence may additionally include **boundary maintenance at P → sustained regime**.

O is not restricted to a conscious human witness. Depending on the system, an observer-positioned registration function may be instantiated by measurement, internal feedback, boundary response, environmental coupling, or another durable state-recording process.

This is an architectural hypothesis, not a claim that human observation creates physical reality or that every measurement causes a LUCY transition. A developed model must identify the relevant registration channel and predict what changes when it is removed, delayed, corrupted, or decoupled.

4. Layer discipline

MKUFT uses the recursive typed architecture

$$S \rightarrow I \rightarrow P \rightarrow O \rightarrow S.$$

The layers are addresses rather than ordinary physical dimensions.

LUCY may be addressed as LUCY(I) for informational or routing constraint formation, LUCY(P) for physical realisation, and LUCY(O) for observer-positioned registration or closure where operationally defined. A cross-layer LUCY claim must additionally name the carrier that takes the relation from one address to another.

A result at one address does not prove a result at another.

No untyped inheritance.

An informational threshold is not evidence of a new physical field. A physical LUCY claim begins only when the informational relation has a defined P-layer consequence that survives comparison with the adequate physical baseline. An observer-state effect cannot become a material metric change without an independently measured P-layer carrier.

4.1 Independent-content requirement for I→P claims

A physical LUCY claim must not be created by arbitrary labels, wording, notation, or human naming alone. The informational object should be specified through relational or structural features that survive appropriate re-encoding.

If the proposed informational variable is fully reducible to the already measured physical state, then the I-layer may remain a useful description or compression, but the experiment has not established independent cross-layer physics.

A sober physics-facing discriminator is therefore:

after the relevant physical state and history are controlled, does the declared informational relation still improve prospective prediction of the P-layer transition or weighting?

If not, the stronger I→P LUCY branch reduces toward the ordinary physical account.

5. Established territory and proposed addition

LUCY does not claim novelty for phase transitions, critical thresholds, metastability, hysteresis, domain walls, interface energy, collective order parameters, symmetry breaking, nuclear shell or pairing effects, general relativity's geometric treatment of gravitation, or the emergence of several observables from one underlying state.

The framework overlaps research on induced and emergent gravity, thermodynamic gravitational dynamics, analogue gravity, many-body spacetime models, scalar-tensor phase transitions, spontaneous scalarisation, vacuum-state engineering, and nonlinear material or plasma boundaries.

The proposed MKUFT contribution is the specific conjunction of typed layer separation, a thresholded cross-layer crossing condition, relational closure, geometry or admissible-path state as a possible selected physical output rather than amplified force, optional membrane and pocket regimes, strongest-fair-null comparison, ordinary-physics recovery, and branch-level falsification.

6. Minimal threshold notation

A domain-specific LUCY model may define a yield index

$$Y_L(x, t) = \chi_L \frac{\|\nabla \tau_L(x, t)\|^2}{N_L(x, t) + \varepsilon_L},$$

with prespecified threshold

$$Y_L(x, t) \geq Y_*.$$

Here τ_L is a declared scalar gradient variable, χ_L a coupling or scale factor, N_L a noise/disorder/competing-load measure, $\varepsilon_L > 0$ a regulariser, and Y_* the preregistered threshold.

This is a template rather than a universal law. It is valid only when the terms are defined, units or normalisation are consistent, and threshold crossing predicts a distinct downstream state or transition.

The threshold is not required to trace a circle or closed spatial surface. Its geometry belongs to the variables and carrier of the active model.

Where curl-like behaviour is required, a vector potential must be introduced. A scalar τ_L cannot acquire a curl by notation alone.

7. State-selection scaffold

Let $q(x, t)$ be an order parameter and λ a set of declared control parameters. A conventional effective functional is

$$\mathcal{F}[q; \lambda] = \int_V \left[\frac{\kappa_q}{2} \|\nabla q\|^2 + V(q; \lambda) \right] dV + \mathcal{F}_{\text{boundary}} + \mathcal{F}_{\text{environment}}.$$

A toy potential may be

$$V(q; \lambda) = \frac{a(\lambda)}{2} q^2 + \frac{b}{4} q^4 + \frac{c}{6} q^6, \quad c > 0.$$

The selected state is

$$q_* = \arg \min_q \mathcal{F}[q; \lambda].$$

A LUCY critical surface Σ_L may occur where competing preferred solutions exchange stability:

$$\Delta \mathcal{F}(\lambda) = \mathcal{F}[q_1; \lambda] - \mathcal{F}[q_0; \lambda] = 0.$$

This captures threshold and competing stable states. Hysteresis, interface formation, membrane formation, and sustained domains are additional regime-specific consequences when the chosen physical model supports them. It does not establish a new physical coupling.

8. Membrane and pocket energetics

This section applies only to LUCY branches that actually predict a bounded physical region.

For a roughly spherical physical pocket of radius R , a schematic nucleation cost is

$$\Delta E(R) = 4\pi R^2 \sigma - \frac{4\pi}{3} R^3 \Delta \varepsilon + E_{\text{load}}(R) + E_{\text{diss}}(R),$$

where σ is interface energy per unit area, $\Delta \varepsilon$ the bulk energy-density difference, and the final terms represent declared load and dissipation costs.

In the simplest uncoupled case,

$$R_c = \frac{2\sigma}{\Delta \varepsilon}$$

is the critical radius above which volume gain can exceed surface cost.

This is standard nucleation grammar used as a discipline for one morphology class. It is not the generic geometry of LUCY. A non-spherical, nonlocal, graph-like, or non-boundary implementation requires its own appropriate cost and stability model. “Coherence” is not free energy.

9. Gravity-like behaviour as output

General relativity already models gravitation through spacetime geometry rather than an ordinary Newtonian force. LUCY does not claim that conceptual move as new.

The open proposal is:

A relational configuration may cross an $I \rightarrow P$ threshold at which it changes physical admissibility or selects a different effective geometry. Matter paths, light paths, clock relations, and inertial behaviour would then be correlated outputs of that physical state.

The proposed chain is **constraint configuration $\rightarrow$ relational closure $\rightarrow$ consequential $I \rightarrow P$ crossing $\rightarrow$ selected effective geometry, where present $\rightarrow$ matter/light/clock observables.**

It is not a claim that a small gravitational wave is amplified into a large gravitational force.

A possible scale relation between a local high-gradient regime and an extended effective-geometry regime would require an explicit typed map,

$$C_{\ell \rightarrow L} : \mathcal{X}_\ell \rightarrow \mathcal{X}_L,$$

showing how microscopic configuration becomes a macroscopic order parameter while preserving causality, energy accounting, and tested equivalence-principle limits. Until then the coupling may be zero.

10. Extreme configuration-sensitive matter

Neutron-rich superheavy nuclei are one candidate seed class because their stability can depend sharply on shell corrections, pairing, deformation, and fission barriers. Astrophysical rapid-neutron-capture environments may reach configurations unavailable in present terrestrial synthesis.

The permitted statement is that such environments may produce configuration-sensitive nuclei whose lifetimes differ substantially from measured neutron-deficient isotopes.

That does not imply altered gravitation. No accepted nuclear model supplies such a coupling, and within LUCY the coupling remains a hypothesis that may be absent.

A nucleosynthesis exposure parameter may be written

$$\Theta_r = \tau_{\text{env}} n_n \langle \sigma v \rangle.$$

This describes neutron-capture exposure. It is not a geometry-changing parameter by analogy.

11. Geometry, scale, and threshold

Scaling can change category rather than merely magnitude, as in laser threshold, superconducting transition, critical nucleation, collective modes, and spontaneous ordering.

A valid scaling claim requires declared extensive and intensive variables, coherence length where relevant, finite-size corrections, geometry and boundary dependence where relevant, dissipation, critical exponent or explicit non-universal law, and ordinary limit.

The geometry of the physical consequence must be derived or measured from the active model. It must not be inherited from the word “local” or from earlier membrane illustrations.

A collective term proportional to N^2 cannot be inserted merely because coherent amplitudes sometimes add quadratically.

12. Discriminating physical signature

The primary physics-facing LUCY signature is a preregistered threshold at which a declared informational or relational variable adds prospective predictive power for a P-layer transition or state after the adequate physical baseline has been controlled.

For a boundary-forming or effective-geometry branch, a stronger signature requires several normally distinct observables to change at the same mapped boundary and control threshold.

Candidate channels include neutral test-mass motion, optical phase and path, clock comparison, fluid convection, electromagnetic propagation, thermal transport, material stress, pressure, and radiation or plasma diagnostics.

For a boundary claim, the load-bearing signature remains:

One boundary, one onset, one control threshold, and one model predicting the correlated residuals.

For a non-boundary LUCY claim, replace “one boundary” with the declared common transition/address relation. A change in only one channel normally favours a domain-specific explanation.

13. Required discriminators

Threshold and hysteresis

The control parameter and onset are specified before observation. A sharp transition or hysteresis may support a phase-change model; smooth power-proportional response normally favours heating, leakage, or ordinary nonlinear response unless another mechanism is derived.

Independent-content test

For an I→P claim, informational variables must add held-out prospective prediction after the relevant physical state and history are controlled. Relation-preserving re-encoding should not change the physical prediction unless the re-encoding itself changes a physical or observer variable.

Common boundary, when claimed

If the model predicts a membrane or bounded domain, independent sensors must locate the same boundary within declared uncertainty.

Geometry dependence

After controlling for cavity modes, stress, thermal transport, mass distribution, and fluid flow, normalised physical geometry should predict threshold or effect class under a declared scaling relation where geometry is part of the model.

Composition and charge

A metric-like residual should couple approximately universally to neutral bodies. Strong dependence on charge, conductivity, magnetic susceptibility, dielectric constant, or surface finish favours conventional interactions.

Wavelength

After plasma and material dispersion are removed, a metric-like optical-path residual should be approximately achromatic. Strong wavelength dependence favours refraction, absorption, scattering, plasma, or resonant material effects.

Clocks and phase

At least two independent timing or phase methods should show a consistent residual before altered effective geometry is claimed.

Conservation

Source energy, heat, radiation, momentum transfer, mechanical reaction, and entropy production remain part of the accounting.

Astrophysical residue

If a particular extreme nuclear configuration is required, network calculations should predict distinctive abundance, decay-chain, or fission-fragment signatures. Failure to find the predicted signature as nuclear data improve weakens that seed branch.

14. Strongest fair null

Every physical test requires the strongest reasonable conventional explanation set, including electromagnetic, electrohydrodynamic, plasma, thermal, acoustic, mechanical, fluid-dynamic, optical, material, sensor/reconstruction, and relevant gravitational alternatives, including combined multifactor models.

For an $I \rightarrow P$ claim, the null must also include an adequate physical-state model and relevant physical history so that omitted P-layer variables are not misclassified as informational dynamics.

The proposed LUCY model carries weight only where it predicts held-out observations better than the strongest fair null under matched information, calibration, and parameter freedom.

15. Evidence and promotion gates

Tier 0 — mathematical consistency

Define spaces, variables, dimensions, update rule, typed carrier, stability, ordinary limit, and causality.

Tier 1 — established analogue reproduction

Demonstrate a preregistered threshold with a consequential downstream state change in a known system without claiming new gravitation or independent cross-layer physics. Membrane and pocket behaviour may be demonstrated where the analogue actually supports those regimes.

Tier 2 — reproducible discriminating residual

Under blinded and sham-controlled conditions, observe the preregistered target-layer residual. For an $I \rightarrow P$ claim, the informational variable must improve held-out prediction beyond the adequate physical baseline. For a boundary claim, locate one mapped boundary in at least two independent channels.

Tier 3 — effective-geometry discrimination

Where altered geometry is claimed, show that one effective geometry predicts neutral matter, light, and clock behaviour better than the strongest conventional alternatives.

Tier 4 — independent replication and scaling

A separate laboratory reproduces the effect and the preregistered threshold, representation-invariance, geometry where relevant, composition, wavelength, and hysteresis predictions.

No foundational or ontology claim is promoted before these gates are met.

16. Ordinary-physics recovery

A developed LUCY model must recover ordinary behaviour when

$$Y_L < Y_*,$$

when the order parameter remains in the baseline minimum, the proposed cross-layer coupling is zero or reducible to the adequate physical state, the target-layer residual falls below declared sensitivity, or the additional model adds no held-out predictive value.

For boundary-forming branches, ordinary behaviour is also recovered when the membrane is unstable or a proposed pocket remains below its critical stability condition.

In those regimes the addressed domain is described by its established ordinary theory.

A deeper physical LUCY account must also recover general relativity and quantum field theory within their tested domains.

17. Falsification and reduction

A LUCY branch is weakened, reduced, or rejected when the threshold cannot be operationally defined; the post-threshold state is not reproducibly distinct; the proposed cross-layer carrier cannot be defined; the informational variable is reducible to an adequate physical-state description; relation-preserving re-encoding changes the claimed physical result without a physical reason; held-out predictive advantage disappears under temporal shuffling or adequate controls where temporal co-registration is claimed; a claimed membrane or common boundary cannot be located; different channels identify incompatible boundaries where one boundary was predicted; effects scale with ordinary apparatus variables sufficient to explain them; the proposed geometry does not outperform the strongest conventional model; a dimensionally valid physical coupling cannot be constructed; conservation accounting fails; claimed scaling is fitted only after the result is known; notation changes meaning across layers; a proposed nuclear or material seed is unnecessary; held-out predictions fail; or the framework adds no discrimination beyond existing formalisms.

A failed physical branch is not rescued by relocating the effect to an unmeasured informational or observer layer.

18. What LUCY is not

LUCY is not a place, bubble, circle, sphere, membrane, pocket, new force by declaration, gravity particle, gravity wave, universal field substance, synonym for consciousness, proof of extra dimensions, licence to treat analogy as mechanism, source of free energy, replacement for general relativity, replacement for quantum field theory, or label for every coherent system.

A membrane or pocket may be a physical consequence of a particular LUCY branch. It is not LUCY itself.

LUCY is a thresholded layer-crossing architecture. Each physical application must earn its own carrier, mechanism, geometry, and evidence.

19. Physics-facing application

The dedicated downstream application is [LUCY Threshold Geometry and Relational Closure](#). That paper explores one boundary-forming effective-geometry branch. Its membrane and pocket morphology is a downstream specialisation and does not replace the more general definition owned here.

20. Related public documents

- [Layer Before Law](#)
- [Cross-Layer Invariants and Layer Addressing](#)
- [Gradient Mechanics and Boundary-Shear Geometry](#)
- [LUCY Threshold Geometry and Relational Closure](#)
- [OCQS Human Activation Layer](#)
- [Typed Traversal and Equation Hygiene](#)
- [Discriminating Experiments and Promotion Gates](#)
- [Falsification Summary](#)
- [Cross-Support and Traversal Map](#)

21. Compact rule

Define the addressed relation. Pre-register the threshold. Define the carrier. Measure the downstream consequence. Map a membrane or sustained pocket only if the physical branch actually predicts one. Test the strongest physical null. Recover ordinary physics. Remove the branch if the discriminator fails.

LUCY is the proposed crossing threshold where an addressed relation becomes consequential at the next layer. LUCY-1 is a membrane regime when a stable interface follows. LUCY-2 is a sustained pocket when that interface supports a finite region. Gravity-like behaviour, where investigated, is treated as a possible downstream output of selected effective geometry rather than a wave, particle, force, or assumed bubble.